

Effect of selective decontamination of the digestive tract of donor and recipient on the occurrence of murine delayed-type graft-versus-host disease

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Abstract. In the present study we investigated the occurrence of delayed-type graft-versus-host disease (DT-GvHD) after allogeneic bone marrow transplantation (BMT) between two H-2 incompatible mouse strains. BMT was performed on mice with a conventional intestinal microflora as well as on mice in which the *Enterobacteriaceae* were selectively eliminated from the intestinal microflora by oral antibiotic treatment. None of the conventional or the selectively decontaminated (SD) chimaeric mice suffering from DT-GvHD died of bacteraemia. While DT-GvHD was mitigated when C3H/He recipient mice were SD-treated, this was not the case when C57Bl/6J recipient mice were SD-treated. SD-treatment of the digestive tract of donor mice only mitigated DT-GvHD when the recipients were also SD-treated. We conclude that *Enterobacteriaceae* in the digestive tract may only play a minor role, if any, in the occurrence of DT-GvHD. Instead, we postulate that in this study DT-GvHD was determined by differences in the composition of the resident intestinal microflora (IM) of both mouse strains together with the cellular composition of the bone marrow graft. The interaction between antigenic components of the recipient's IM and the developing donor immune system in the recipient as a possible cause for DT-GvHD is discussed.

Introduction

Bone marrow transplantation (BMT) has become a well-established therapy for leukaemia, aplastic anaemia and severe combined immuno-deficiency disease in man. However, acute or delayed-type graft-versus-host disease (GvHD) may occur as a serious complication after BMT. Acute GvHD is generally believed to occur as a result

of the reaction of cytotoxic T-lymphocytes from an immunogenic unrelated donor to tissue antigens of the recipient. The occurrence of acute GvHD can thus be markedly reduced after T-cell depletion of the bone marrow graft [7, 18].

Delayed-type (DT) GvHD in mice occurs from 3 weeks after BMT. Mice suffering from DT-GvHD may show wasting, diarrhoea and skin lesions as clinical symptoms, whereas organs like the liver, gut, lungs, skin, spleen and thymus may show characteristic lesions [19, 30]. Animal and human studies have shown that donor T-lymphocytes do not play an important role in the occurrence of DT-GvHD [7, 18, 21]. However, it has been postulated that precursor T-lymphocytes can mature in the thymus of the recipient and subsequently play a role in DT-GvHD [18, 21].

More comprehensive evidence is available about the role of the gastro-intestinal microflora of the recipient in the occurrence of DT-GvHD as germfree mice do not develop DT-GvHD when engrafted with allogeneic bone marrow [14, 22]. Absence of DT-GvHD was also observed when the digestive tract of recipients had been selectively decontaminated (SD) from *Enterobacteriaceae* [23].

The interaction between the developing allogeneic immune system and the recipient's microflora in DT-GvHD still has to be elucidated. At least three factors, concerning *Enterobacteriaceae* in the digestive tract of the recipient should be taken into account:

- (1) Infections by *Enterobacteriaceae* may be an important cause of death in a state of immunological imbalance during DT-GvHD [18, 20]. SD-treatment of the recipient may therefore influence the survival after allogeneic BMT;
- (2) *Enterobacteriaceae* appear to be major contributors to endotoxin in the murine gut [10]. Translocation of endotoxin may be enhanced after total body irradiation due to mucosal lesions or an impaired immunological defence. As endotoxin has been found to stimulate graft versus host reactivity [18], reduction of the intestinal endotoxin concentration may possibly be important to mitigate DT-GvHD;
- (3) It has been postulated that *Enterobacteriaceae* may share antigens with tissue antigens of the recipient [21]. Because of this cross reactivity, such bacteria may enhance the reactivity of allogeneic donor lymphocytes. However, hypothetical cross-reacting microorganisms appear to induce DT-GvHD up to only 40 days after BMT, since DT-GvHD has been reported not to occur if the microflora is absent in the recipient up to 40 days after BMT [23].

The effect of SD-treatment on the recipient has been studied, but no attempt has yet been made to determine the effect of treating the donor in this way. Recently it has been found that SD of the digestive tract may increase the proliferation state of bone marrow stem cells and precursor cells [10, 11]. This might be of particular interest, since it is well known that the cellular composition of marrow graft also determines the degree of GvHD [7, 18, 21]. Subsequently, a change in the number of proliferating precursor T-lymphocytes in the BM graft may influence the occurrence of DT-GvHD.

The present experiments were performed to study the effect of SD-treatment, that is, the elimination of *Enterobacteriaceae* from the digestive tract [20, 25, 27] of donor and/or the recipient on the occurrence of DT-GvHD after allogeneic BMT. As *Enterobacteriaceae* contain endotoxin and endotoxin may influence DT-GvHD, the

quantity of this was measured in the faeces. Aerobic cultures of faeces were performed to control the effect of oral administration of antibiotics for SD. All recipient mice were screened for DT-GvHD up to 100 days after BMT. Aerobic cultures of heart blood of moribund mice were performed to detect bacteraemia.

Materials and methods

Mice. Conventionally raised male and female C57Bl/6J (H-2^b) and C3H/He (H-2^k) mice were purchased from the Zentral Institut für Versuchstierkunde in Hannover (FRG) and used at 10–14 weeks of age. The body weight varied between 20 and 25 grams per mouse. Both mouse strains are known to be normal responders to endotoxin.

Housing. All animals were housed in conventional animal rooms at the Central Animal Laboratory of the University of Groningen (The Netherlands) and fed with rodent pelleted food (Hope Farms ind., Warmond, The Netherlands) and tap water ad libitum.

Selective decontamination (SD) of the digestive tract. SD-treatment of donor and recipient mice was performed by supplying aztreonam (donated for the study by Squibb, Princeton, NY, USA) in the drinking water (100 mg/l). Treatment of both donors and recipients was started 2 weeks before Bone Marrow Transplantation (BMT), and continued in the recipients for 100 days after BMT [14, 16].

Bacteriological sampling and culturing. One week before BMT, fresh faeces of conventional and SD donors and recipients was collected and pooled per experimental group. The faeces were suspended (1:9 w/v) and serially diluted in 10-fold dilution steps in brain heart infusion (BHI) broth. After overnight incubation at 37 °C, all dilutions were subinoculated into different media to enable logarithmic enumeration of *Enterobacteriaceae*, enterococci, and yeasts. The detection level was 10¹ m.o. per gram faeces. All *Enterobacteriaceae* were typed and identified by the Analytical Profile Index (API) 20E system (Analytab Products Industry, France).

Each moribund mouse was killed for culturing of heart blood in BHI and after overnight incubation at 37 °C, the culture was subinoculated onto different media for identification.

Determination of the intestinal endotoxin concentration. The intestinal endotoxin concentration was measured in the faeces of two groups of five animals which were randomly chosen from C57Bl/6J and C3H/He mice not participating in the BMT experiments. The intestinal endotoxin concentration before and during SD-treatment was determined by a microtechnique [8] of the limulus amoebocyte lysate assay [16] as used and described by Goris et al. [9].

Micromorphology of the faeces. Faecal samples of both mouse strains used for the endotoxin concentration assay were Gram-stained and screened microscopically for morphological differences in the composition of the faecal floras.

Bone marrow transplantation. Conventional (Conv) recipient and SD-treated recipient mice were placed in perspex irradiation cages and lethally irradiated (9 Gy.) (Philips Mueller MG300 X-ray machine) (200 kV; 15 mA; filters: Cu 0.5 mm/Al 0.6 mm; Rad. rate: 0.45 Gy/min.)

Transplantation experiments were carried out vice versa between C3H/He and C57Bl/6J mouse strains in four, sex matched combinations: Conv → Conv, Conv → SD, SD → Conv and SD → SD. In each experiment about 40 C57Bl/6J or C3H/He were used as recipients and about 20 C3H/He or 15 C57Bl/6J mice as donors.

Bone marrow cell suspensions were obtained by flushing the femurs with Alpha Modification of Eagle's Medium (Flow lab U.K.) plus 1 mM Hepes (N-2 hydroxy ethylpiperazine-N'-2-ethanesulphonicacid) pH 7.2 (Flow lab U.K.). Cell suspensions of 5 to 20 mice were pooled; 24 h after irradiation, each recipient mouse was injected with 1.0×10^7 nucleated cells into a tail vein.

On days 80 and 100, the investigation of chimaerism was performed by haemoglobin electrophoresis accordingly the Gelman haemoglobin electrophoresis method (Gelman Instrument Company, Michigan, USA).

Detection of DT-GvHD. a) Clinical signs. Up to 100 days after irradiation, the body weights of recipient mice were determined weekly. A persistent decrease in body weight of more than 10% of the starting weight, displayed between 21 and 100 days after irradiation, was recorded as loss of (body) weight. During this period, recipient mice were also screened daily for diarrhoea and skin lesions as clinical features of DT-GvHD.

b) Survival rate assay. Only recipient mice that died or were moribund 21 days or later after irradiation were recorded for the survival rate assay. Moribund mice showing a combination of severe weight loss, hunched back, strong decrease in mobility or severe skin lesions were killed by cervical dislocation for bacteriological and histological examination.

c) Histology. Spleen, skin, liver, colon, ileum and lung tissue from moribund mice was sampled. All tissues were fixed in Zenker's solution, hematoxylin/eosine and brachet stained and were then examined microscopically for histological features of GvHD [22, 30].

Statistical analysis. Statistical evaluation of the significance of differences between male and female, conventional and SD-treated mice in the survival rate assay and in the occurrence of clinical symptoms (in the surviving (surv) and nonsurviving (non-surv) animals) was performed by the chi-square test. The significance level was taken at $P < 0.05$.

Results

Results of bacteriologic culturing

a. Faecal cultures. The results of aerobic culturing of faecal samples of conventional (Conv) and SD-treated donor and recipient mice are shown in Table 1. After 1 week

Table 1. Concentration of aerobic micro organisms per gram faeces in conventional (Conv) and Selectively Decontaminated (SD) male (M) and female (F) C57Bl/6J and C3H/He mice. SD was performed by continuous oral aztreonam treatment in the drinking water (100 mg/l). After one week of aztreonam treatment no *Enterobacteriaceae* species could be detected in the faeces of both strains. Isolated *Enterobacteriaceae* species were biotyped by the API 20E system classification method

mice			aerobic micro organisms per gram faeces					
			E. coli 5144572	E. coli 5144532	Ent. cloacae 3305773	Pr. mira- bilis 0736000	Str. faecalis	Yeasts
M	Conv	C57Bl/6J	10 ⁵	—	10 ³	—	10 ⁵	—
		C3H/He	10 ⁴	10 ³	—	10 ⁴	10 ⁵	—
	SD	C57Bl/6J	—	—	—	—	10 ⁵	—
		C3H/He	—	—	—	—	10 ⁵	—
F	Conv	C57Bl/6J	10 ⁴	—	—	10 ⁵	10 ⁵	—
		C3H/He	10 ³	—	—	10 ⁴	10 ⁴	—
	SD	C57Bl/6J	—	—	—	—	10 ⁵	—
		C3H/He	—	—	—	—	10 ⁵	—

of oral-aztreonam-treatment no *Enterobacteriaceae* could be cultured in the faeces of C3H/He as well as C57Bl/6J mice. All SD-treated recipient mice remained free of *Enterobacteriaceae* during the course of the experiment.

b. Heart blood cultures of moribund mice. Aerobic heart blood cultures of killed moribund Conv mice merely showed *Enterobacteriaceae* species like *Escherichia coli*, *Proteus mirabilis* and *Enterobacter cloacae* occasionally in combination with *Streptococcus faecalis* up to two weeks after BMT. The *Enterobacteriaceae* were of the same biotype as the species that had been isolated from the faeces previously, thus were endogenous in origin. In the aerobic heart blood cultures of moribund SD-treated mice, which were killed up to 2 weeks after BMT, only *Str. faecalis*, *St. epidermidis* or *St. aureus* were isolated. However, all the aerobic heart blood cultures of the killed moribund mice were negative, starting from the third week after BMT.

Micromorphological changes in the faecal floras

Between the faecal microfloras of Conv and SD-treated C3H/He mice no micromorphological changes were observed. However, between the faecal microfloras of Conv and SD C57Bl/6J mice a moderate but obvious decrease was seen in the number of fusiform bacteria, together with an increase in rod shaped and coccoid bacteria (data not shown).

Intestinal endotoxin concentration

No significant difference in the intestinal (faecal) endotoxin concentration was detected between Conv and SD-treated C57Bl/6J and C3H/He mice. The mean endotoxin concentration in all groups remained 10 $\mu\text{g/g}$ faeces; this was low at the onset compared with previous experiments of Goris et al. [10].

DT-GvHD related symptoms

The survival rate assay, the occurrence of diarrhoea and the loss of body weight as DT-GvHD related features are shown in Figs. 1–3. In none of the results on survival-rate and clinical symptoms any significant difference was found between comparable sex-matched transplanted male and female groups.

a) *Survival rate assay.* The survival rate curves are presented in Figs. 1 and 2. Whereas SD-treatment of recipient mice did not significantly change the survival rate in C57Bl/6J recipients (Fig. 2A), it did increase the survival rate in C3H/He recipients after day 52 (Fig. 1A).

SD-treatment of C3H/He donors did not change the survival rate in Conv C57Bl/6J recipients compared with the Conv $\rightarrow$ Conv group (Fig. 2B). However, the survival

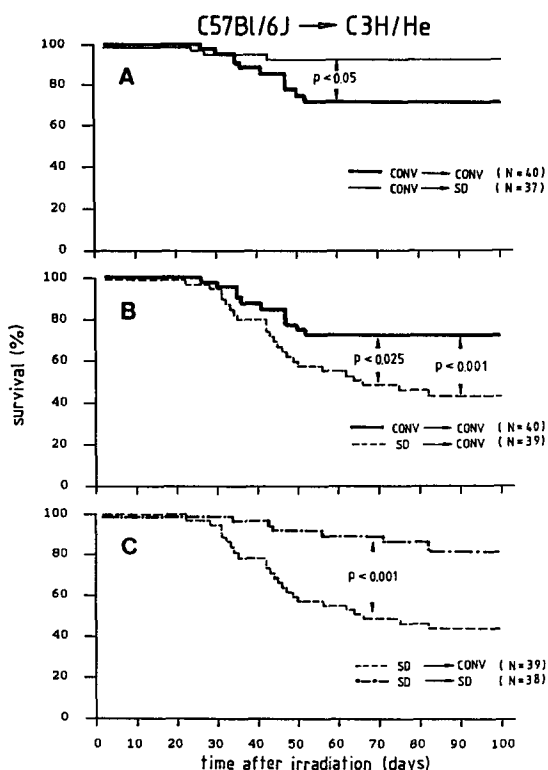


Fig. 1A–C. Survival curves of irradiated (9 Gy) Conventional (Conv) and Selectively Decontaminated (SD) C3H/He (H-2^k) mice from 21 till 100 days after irradiation. Recipient mice were injected intravenously with 1.0×10^7 bone marrow cells from Conv or SD C57Bl/6J (H-2^b) mice. SD-donor and -recipient mice were continuously treated with aztreonam (100 mg/ml) in the drinking water, starting two weeks before transplantation. All SD mice became permanently free of *Enterobacteriaceae* after one week of aztreonam treatment

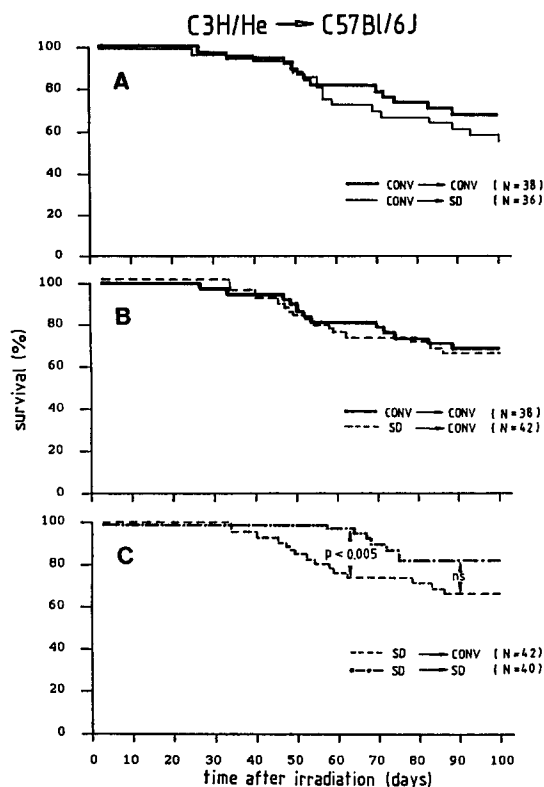


Fig. 2A–C. Survival curves of irradiated (9 Gy) Conventional (Conv) and Selectively Decontaminated (SD) C57Bl/6J (H-2^b) mice from 21 till 100 days after irradiation. Recipient mice were injected intravenously with 1.0×10^7 bone marrow cells from Conv or SD C3H/He (H-2^k) mice. SD-donor and -recipient mice were continuously treated with aztreonam (100 mg/ml) in the drinking water, starting two weeks before transplantation. All SD mice became permanently free of *Enterobacteriaceae* after one week of aztreonam treatment

rate in Conv C3H/He mice decreased after engraftment with SD C57Bl/6J bone marrow (Fig. 1B).

The decreasing effect of SD-C57Bl/6J murine bone marrow on the survival rate in Conv C3H/He recipients was mitigated when the latter were SD-treated as well (Fig. 1C). Mortality was clearly delayed in SD-treated C57Bl/6J recipients after engraftment with bone marrow cells from SD-treated C3H/He donors. However, the difference in the survival rate was not significant on day 100 after BMT (Fig. 2C).

b) Clinical signs. The occurrence of diarrhoea and loss of weight in surviving (surv) and non-surviving (non-surv) recipients were compared with the percentage of non-surv mice (mortality) (Fig. 3).

About half of the Conv → Conv transplanted C3H/He recipients showing diarrhoea and loss of weight died during the experiment, whereas diarrhoea and loss of weight were absent in SD-treated C3H/He mice engrafted with Conv C57Bl/6J bone marrow. The mortality rate was as high as the occurrence of diarrhoea and of weight loss in the SD → Conv transplanted C3H/He recipients, whereas the overall occurrence was not significantly different from that in the Conv → Conv group. The aggravation of clinical symptoms, shown as an increase in mortality, in SD → Conv C3H/He recipients was markedly reduced after recipients had been SD-treated as well (Fig. 3A).

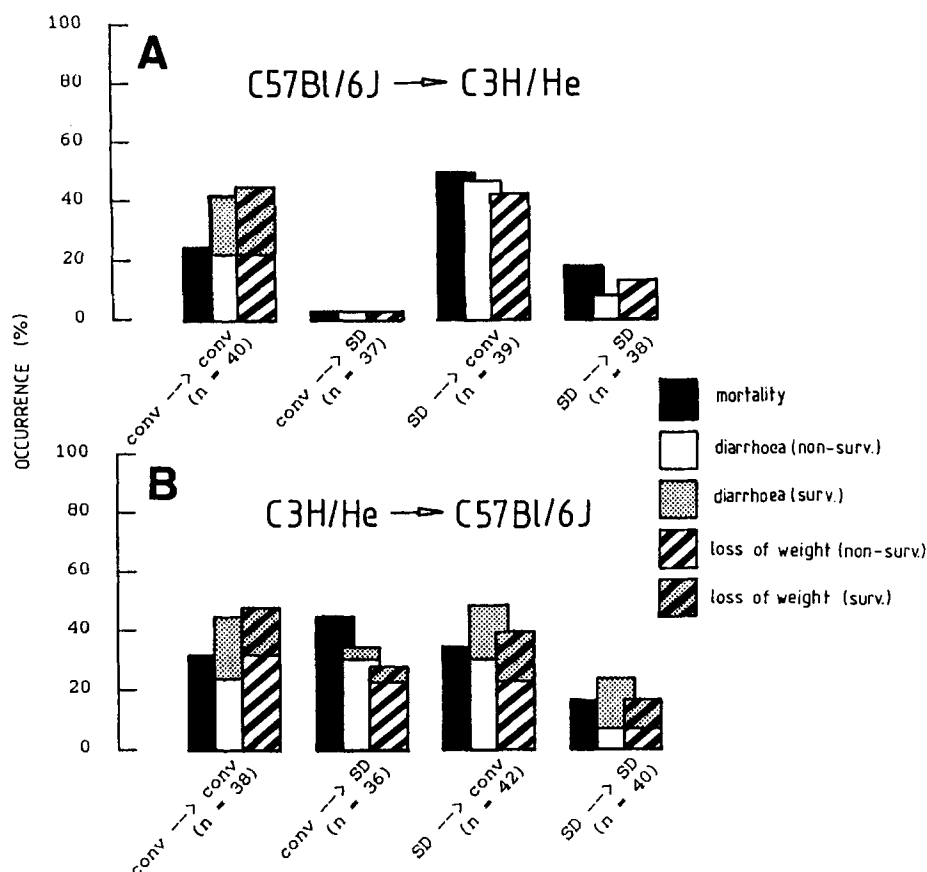


Fig. 3A, B. The occurrence of DT-GvHD related clinical features between 21 and 100 days after irradiation (9 Gy). Conventional (Conv) and Selectively Decontaminated (SD) C3H/He (A) and C57Bl/6J (B) recipients were transplanted with 1.0×10^7 bone marrow cells from Conv or SD donors. SD-donor and -recipient mice were continuously treated with aztreonam (100 mg/ml) in the drinking water, starting two weeks before transplantation. All SD mice became permanently free of *Enterobacteriaceae* after one week of aztreonam treatment. The occurrence of diarrhoea and weight loss is cumulated for mice that survived (surv) and mice that died (non-surv). Significant differences: **A** C3H/He recipients; diarrhoea: (conv → conv)/(conv → SD) $P < 0.005$; (conv → conv)/(SD → SD) $P < 0.005$; (SD → conv)/(conv → SD) $P < 0.005$; (SD → conv)/(SD → SD) $P < 0.005$; loss of weight: (conv → conv)/(conv → SD) $P < 0.005$; (conv → conv)/(SD → SD) $P < 0.005$; (SD → conv)/(conv → SD) $P < 0.005$; (SD → conv)/(SD → SD) $P < 0.005$; **B** C57Bl/6J recipients; diarrhoea: (SD → conv)/(SD → SD) $P < 0.025$; loss of weight: (conv → conv)/(SD → SD) $P < 0.025$; (SD → conv)/(SD → SD) $P < 0.025$

In contrast to C3H/He recipients, diarrhoea, loss of weight and even mortality in the C57Bl/6J recipients was not significantly different in the Conv → Conv, Conv → SD and SD → Conv transplanted groups (Fig. 3B). Only in the SD → SD transplanted C57Bl/6J recipients the occurrence of diarrhoea and of weight loss was significantly lower compared to the SD → Conv group, although mortality was not significantly reduced at day 100 (Fig. 3B). Except for the Conv → SD group, the

occurrence of diarrhoea and of weight loss considered together more or less exceeded the mortality rate; this was so in the Conv → Conv, SD → Conv as well as SD → SD transplanted C57Bl/6J recipients. Mortality exceeded the overall occurrence of diarrhoea and of weight loss only in the Conv → SD transplanted C67Bl/6J recipients (Fig. 3B).

Discussion

The present study shows a distinct effect of the intestinal microflora (IM) on the occurrence of delayed-type graft-versus-host disease (DT-GvHD) after transplantation of allogeneic bone marrow between the two H-2 incompatible mouse strains employed.

The role of Enterobacteriaceae in DT-GvHD. In former studies it has been postulated that *Enterobacteriaceae* in the digestive tract of recipients enhance DT-GvHD after allogeneic BMT [12, 21, 23]. Since elimination of *Enterobacteriaceae* from the digestive tract by oral aztreonam treatment clearly mitigated DT-GvHD in the C3H/He recipients but not in the C57Bl/6J recipients, it seems likely that the occurrence of DT-GvHD is not determined by *Enterobacteriaceae* in the recipient only.

Intestinal endotoxin concentration. The concentration of endotoxin in the intestinal tract may not have played a role in the experiments described because in both aztreonam-treated and conventional mouse strains its concentration in the faeces was at the same low level as in selectively decontaminated mice used by Goris et al. [9, 10, 11]. In the latter studies Conv mice were artificially orally contaminated with an *E. coli* strain. This was not attempted in the present study and may therefore explain why the intestinal endotoxin concentration remained unchanged.

Negative heart blood cultures. Since heart blood cultures of moribund mice were found to be positive only in the first 2 weeks after BMT, bacteraemia could not have been the cause of death in mice showing DT-GvHD.

SD treatment of the recipient. To explain the influence of the intestinal microflora (IM) of recipients and donors on the occurrence of DT-GvHD, it may be useful to divide the resident intestinal microflora (IM) of either recipient strain into an aztreonam-sensitive and -insensitive part. In the C3H/He recipients the aztreonam-sensitive part of the resident IM (including *Enterobacteriaceae*) was found to be associated with DT-GvHD. However, in C57Bl/6J recipients the DT-GvHD associated flora was in the aztreonam-insensitive part of the IM (not containing *Enterobacteriaceae*). Thus DT-GvHD may be mitigated after DT-GvHD associated bacteria have been eliminated by modulation of the IM of the recipient. It may be postulated that if the eliminated fraction involves bacteria which are for some time recognised by an immature or impaired immune system DT-GvHD develops. This hypothesis possibly explains the mitigating effect of aztreonam treatment of the recipient on DT-GvHD in C3H/He mice and the absence of such an effect in the C57Bl/6J mice.

Strain specificity of the Intestinal Microflora (IM). The postulation that aztreonam-sensitive or -insensitive parts respectively influence DT-GvHD in the C3H/He or C57Bl/6J mouse strains, indirectly supports the suggestion that the indigenous intestinal microfloras of both strains differ in composition. A micromorphological difference between the faecal floras of both strains was indeed observed after aztreonam treatment.

On the basis of studies in man and mice it has been postulated that the indigenous part of the IM is related to the genetic characteristics of the host [2, 5, 6, 24]. The composition of the IM may therefore be regarded as strain specific.

SD-treatment of the donor. SD-treatment of donors increased DT-GvHD associated mortality in conv C3H/He recipients. However, no change was seen in the overall occurrence of clinical symptoms (Fig. 3A). It can therefore be concluded that only the degree of DT-GvHD was aggravated. These findings can be explained by the assumption that SD-treatment may have had an effect on the cellular composition of the BM-graft [9, 11], which may determine the development of the (impaired) immunological regulation of the recipient's IM after BMT and in turn induce DT-GvHD.

Impaired immunity after BMT. The thymus and the gut associated lymphoid tissue (GALT) appear to play an important role in the complex mechanism of IM-regulation [26, 28]. Congenitally athymic and newborn thymectomised mice, conventionally maintained, not only suffer from symptoms resembling DT-GvHD [1, 29], but also fail to control the concentration and the presence of bacteria (and other micro-organisms) in the gut [13, 26]. It is therefore possible that the mechanism by which the IM influences DT-GvHD is related to a failure of the engrafted BM (lymphoid-precursor) cells to adapt to the IM of the recipient.

Previous BMT experiments in SD-treated recipient mice revealed that a BM-graft of 10^7 nucleated cells requires some 30 to 40 days to develop to sufficient maturity for conventionalisation without ensuing DT-GvHD [23]. The engrafted (immature) bone marrow cell lineages may therefore, fail for some time to respond properly to components of the recipient's microflora.

After SD-treatment of the donor, the BM graft may consist of relatively few primed lymphocytes. Thus, an instant takeover of the immunological defence against bacteria in the gut may fail. Bacteria or bacterial components may increasingly translocate through the mucosal wall and deposit in various tissues, such as the intestines, liver and skin. Evidence for this hypothesis has been reported in experimental BMT studies in mice as well as in man with DT-GvHD [3, 4]. After development of the donor's immune system in the recipient, impaired local and systemic immunological reactions initiated by bacterial components may result in pathophysiological changes as are seen in DT-GvHD. The degree of DT-GvHD may be aggravated after engraftment of BM from an SD-treated donor, if DT-GvHD associated flora components are present in the recipient; this was seen in SD $\rightarrow$ Conv C3H/He recipients (Figs. 1B and 3A). It is possible that no change in translocation took place in SD $\rightarrow$ Conv C57Bl/6J because no increase in the occurrence of DT-GvHD was found in these mice (Figs. 2B and 3B).

SD-treatment of donor and recipient. Whereas SD-treatment of C57Bl/6J donors aggravated DT-GvHD in Conv C3H/He recipients, DT-GvHD was significantly mitigated when the C3H/He recipients were SD-treated as well. However, there was no reduction in the occurrence of DT-GvHD in the Conv → SD C3H/He recipients. This suggests that SD-treatment of the donor does not mitigate DT-GvHD in recipients with a SD-sensitive DT-GvHD associated flora. However, DT-GvHD was delayed in C57Bl/6J recipients after SD-treatment of both recipient and donor (Fig. 2C). Therefore, despite a possible deficiency in the regulation of the recipient's IM shortly after BMT, lymphocytes of the SD-treated donor may find more optimal circumstances to develop proper immunity to (formerly) DT-GvHD associated flora components in the recipient's IM. Bone marrow transplantation from C57Bl/6J donors into C3H/He recipients support the hypothesis that SD-treatment of donor animals may be important to mitigate DT-GvHD (Figs. 1E and 3A).

In conclusion. The results of this investigation suggest that: 1) *Enterobacteriaceae* and the intestinal endotoxin concentration are of minor importance in DT-GvHD in mice; 2) SD-treatment by nonabsorbable antibiotics may eliminate a DT-GvHD associated part of the intestinal microflora. Such a modulation of the IM may be of primary importance when one attempts to mitigate DT-GvHD. However, in cases in which this cannot be accomplished by SD-treatment of the recipient, SD-treatment of the donor appears to be helpful, as it may reduce the occurrence of DT-GvHD symptoms as well as mortality. Further investigation of the immunological reactivity against bacteria of the indigenous IM will be needed to predict the occurrence of DT-GvHD in SD-treated recipients after allogeneic BMT.

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